

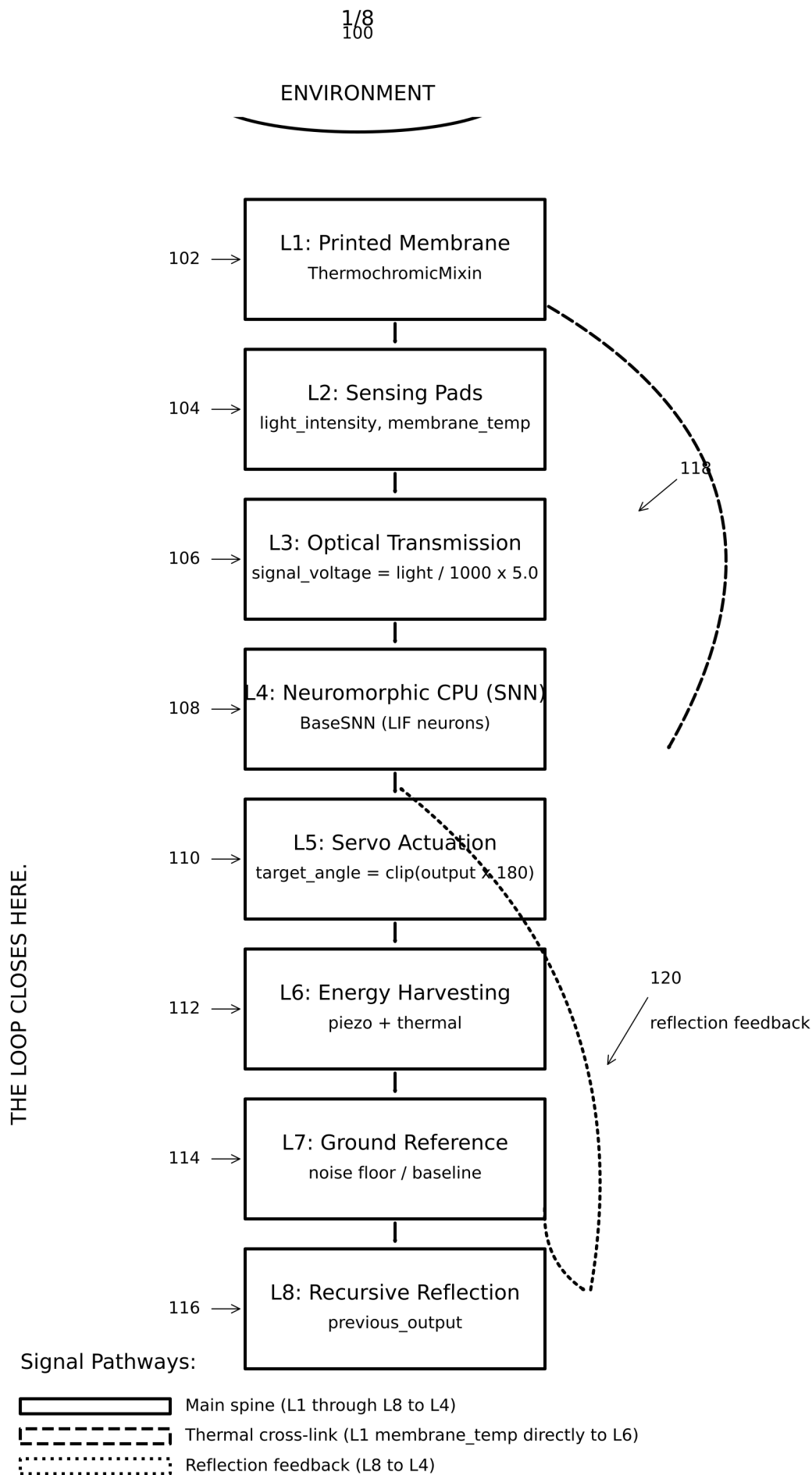


FIG. 1

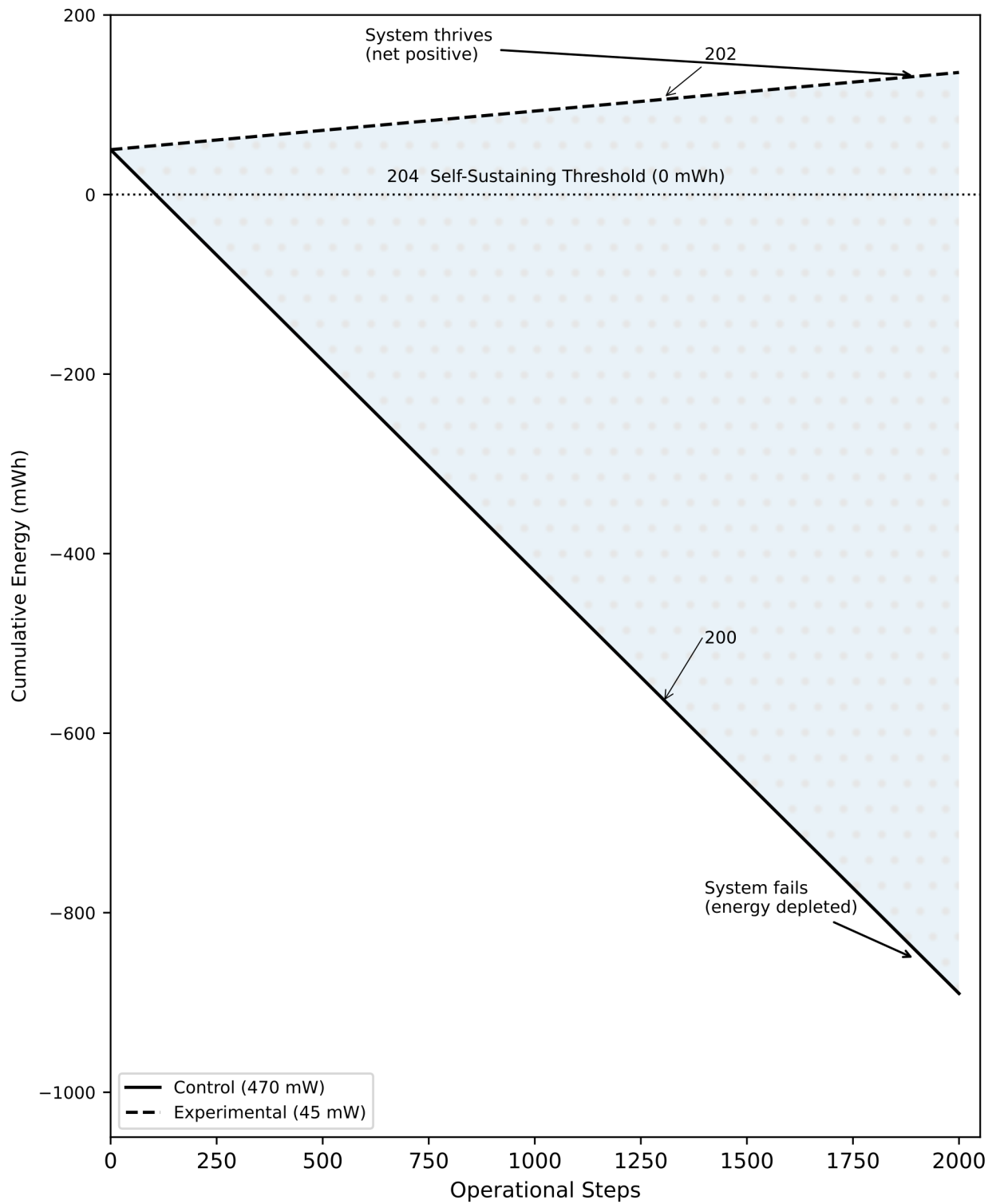


FIG. 2

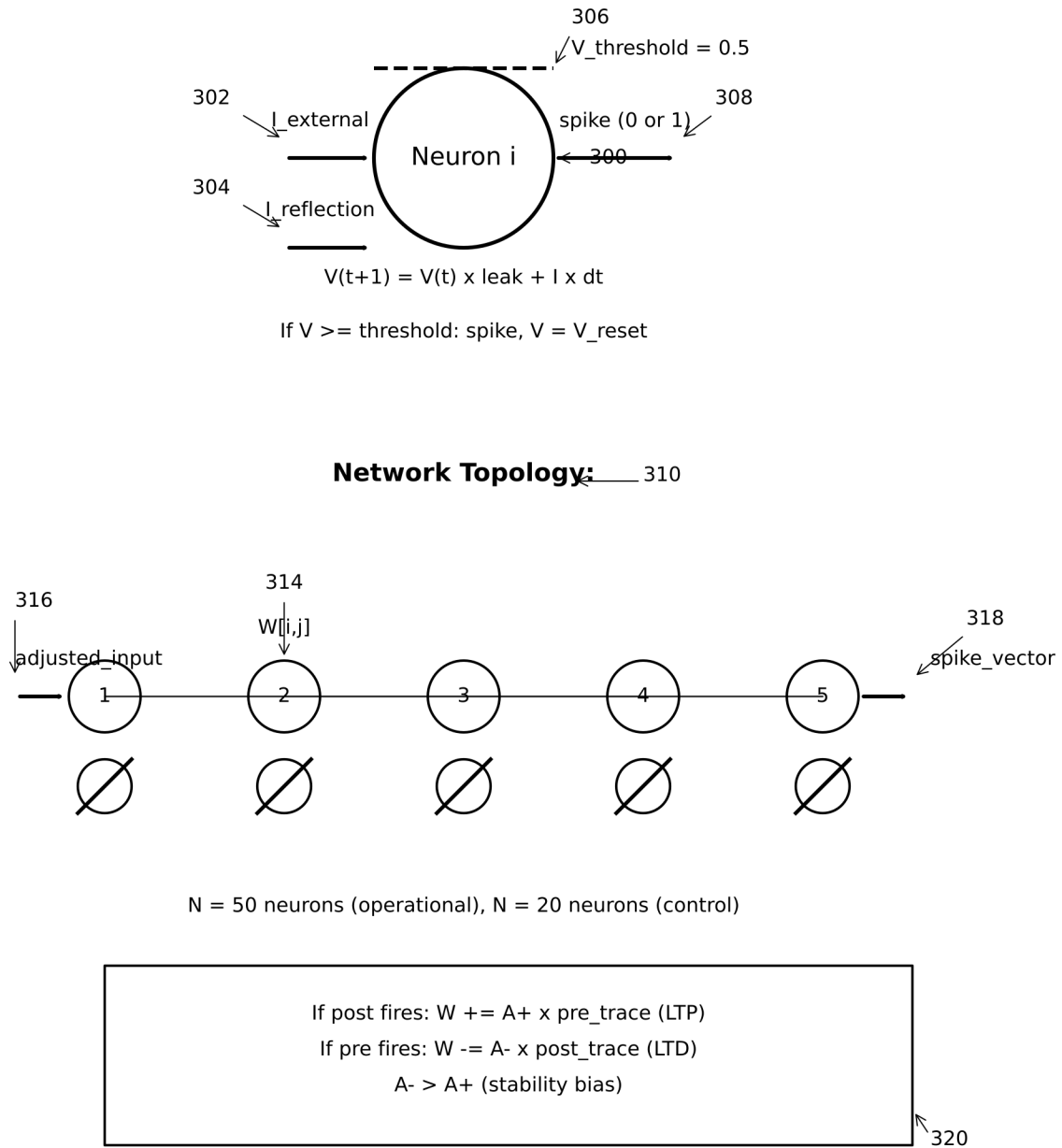


FIG. 3

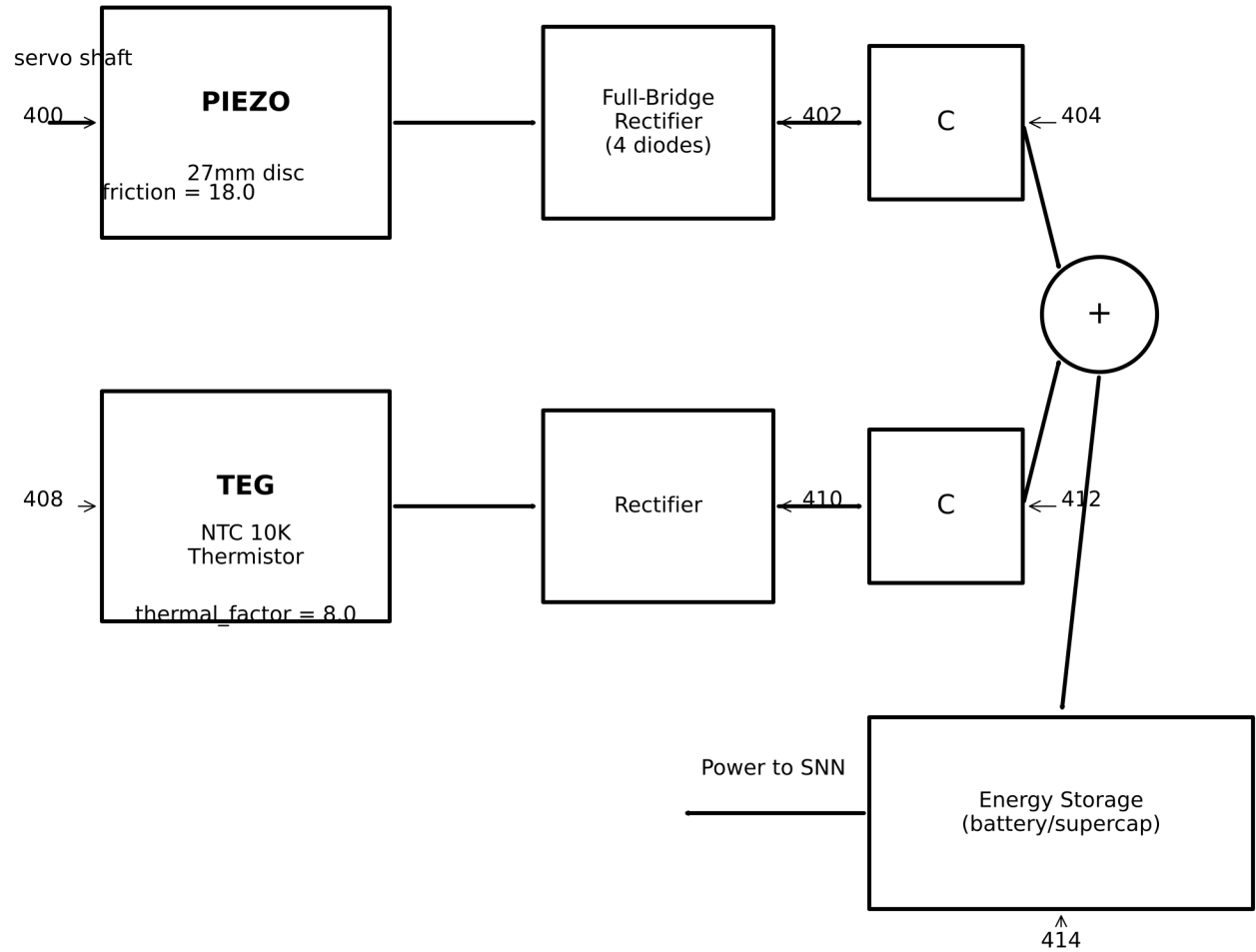


FIG. 4

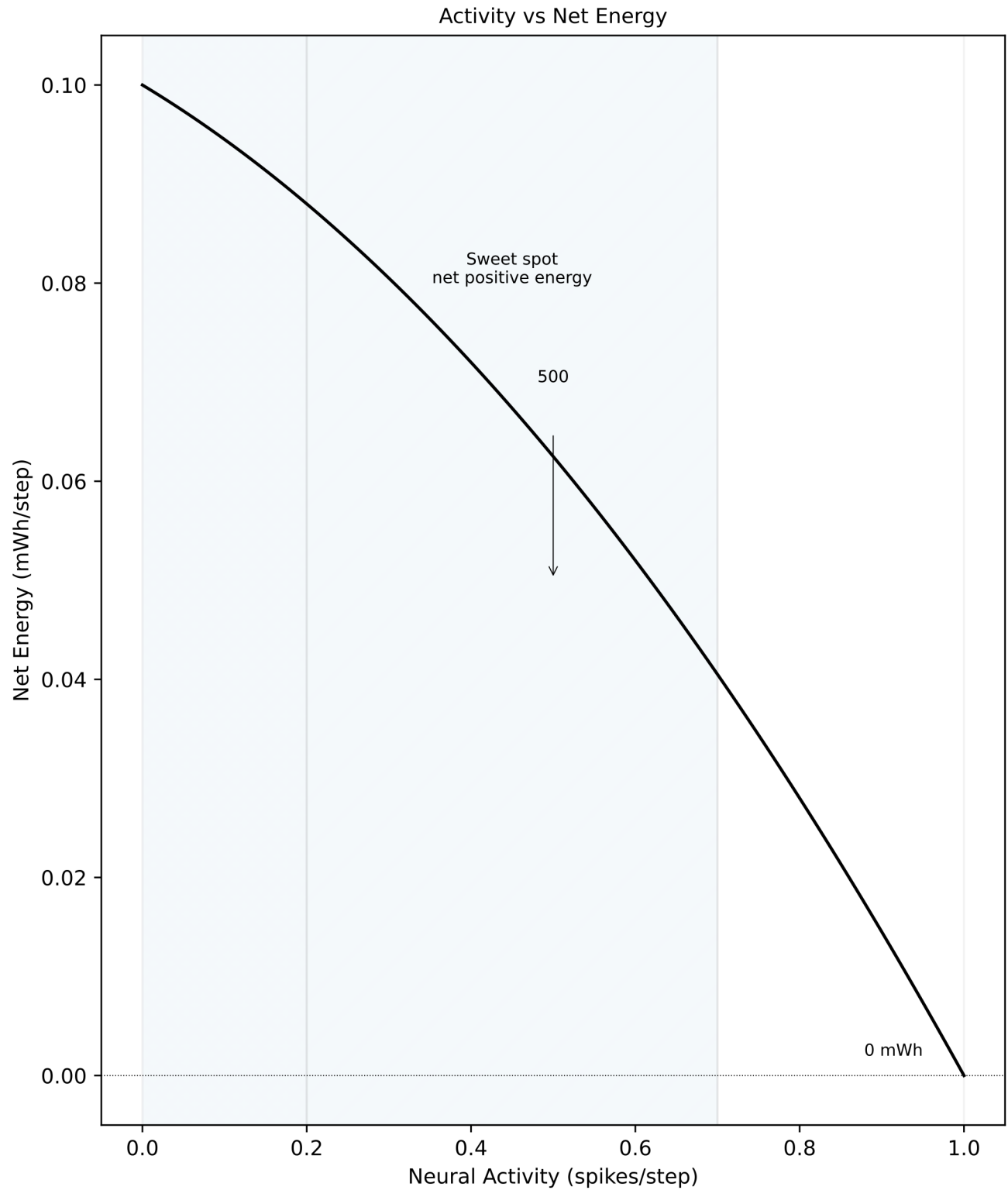
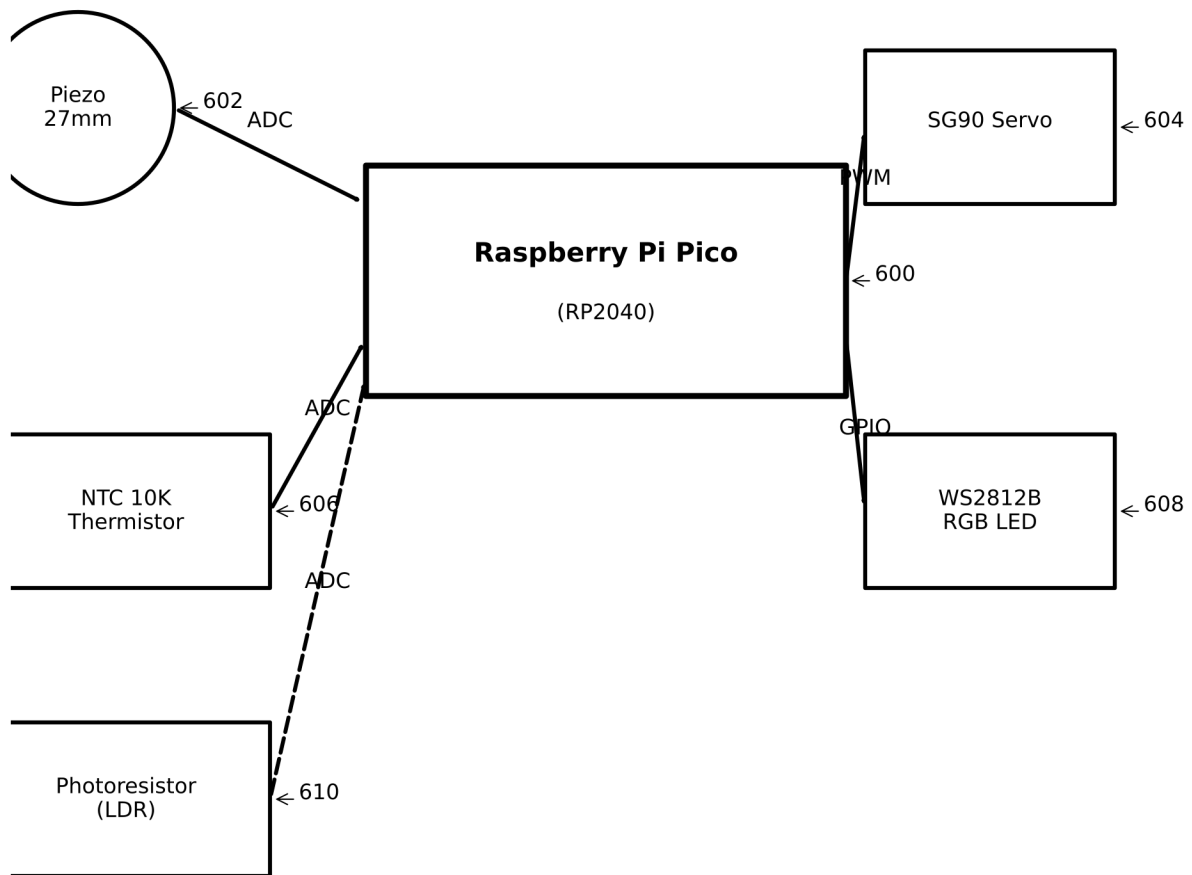


FIG. 5

Low activity
insufficient
harvesting

High activity
quadratic cost
dominates

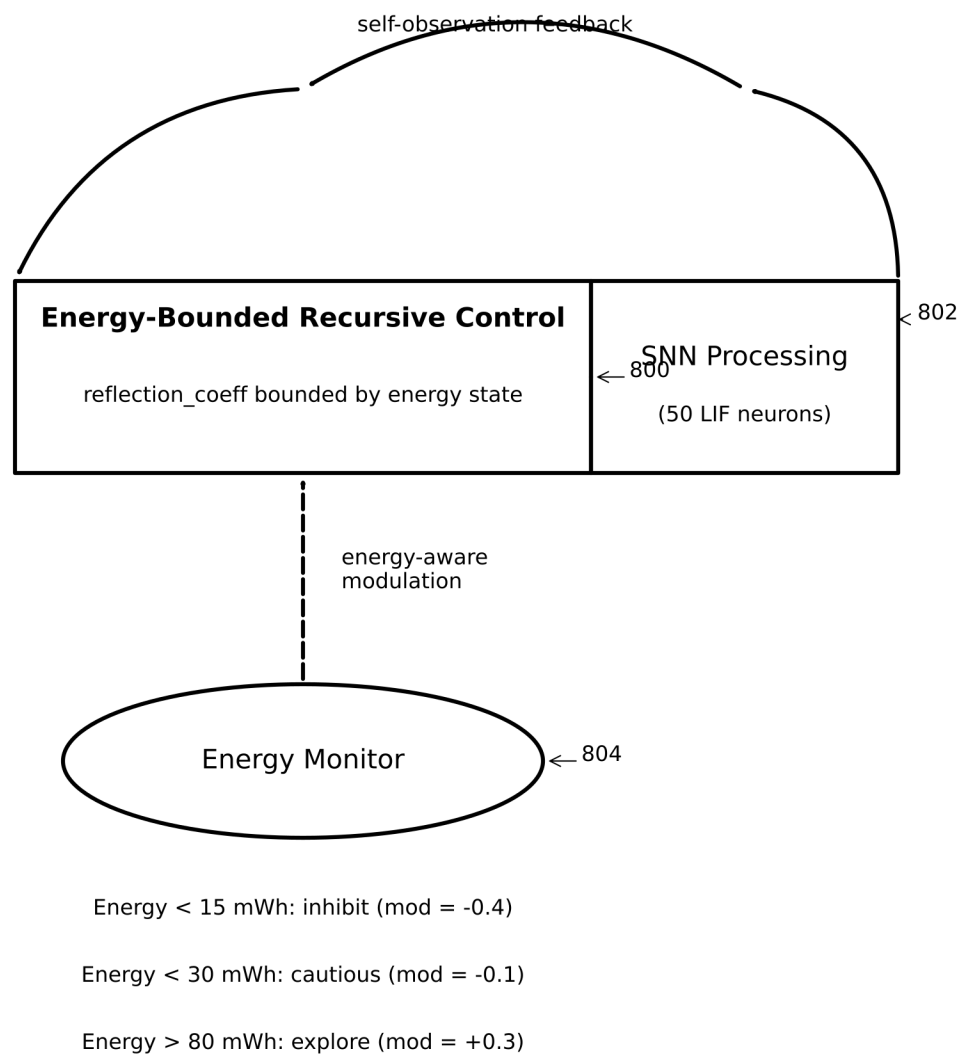


Bill of Materials	
Raspberry Pi Pico	\$4
SG90 Micro Servo	\$3
27mm Piezo Disc	\$2
NTC 10K Thermistor	\$1
Photoresistor (LDR)	\$1
WS2812B RGB LED	\$1
Misc (wires, board)	\$3

Total BOM Cost: <\$25

FIG. 6

Claim	Criterion	Measured	Threshold	Result
	Phase 7 Control Baseline (EnergyConfig)			
A	Closed-loop continuity	2000 pts	2000	PASS
B	Boundedness	bounded	bounded	PASS
C	Robustness (noise)	$\text{Std}(\theta) \leq 45^\circ$	$\leq 45^\circ$	PASS
D	Input-output gain	$\text{corr} \geq 0.2$	≥ 0.2	PASS
E	Saturation ratio	$\text{sat} \leq 0.20$	≤ 0.20	PASS
	Phase 7 Full Integration (Control)			
--	Energy at 2000 steps	-4,697 mWh	< 0	CONFIRMED
	MCP Integration (BalancedEnergyConfig)			
--	Energy at 2000 steps	+4,170 mWh	> 0	CONFIRMED
	Phase 8 STDP			
F8.1	Weight entropy decrease	2.95→2.02	decrease	PASS
F8.2	STDP MI $> 1.2\times$ frozen	6.52x	> 1.20	PASS
F8.3	Weight convergence	0.00%	$< 10\%$	PASS



Recursive control loop bounded by energy levels to ensure self-sustainability.

FIG. 8